

Geophysical Analysis Report

Seismic Agent Multi-Agent System

Mission ID: mission_1232ab0d

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Mission Objective

Analyze the Marmousi synthetic data for porosity prediction.

Data Sources

- MODEL_S-WAVE_VELOCITY_1.25m.segy.tar.gz
- MODEL_P-WAVE_VELOCITY_1.25m.segy.tar.gz
- Vs.segy.tar.gz
- Vp.segy.tar.gz

Quantitative Results

Metric	Value
velocity_files_loaded	['vp', 'vs']
vp.min	1028
vp.max	4700
vp.mean	2672
vp.std	973.8
vp.p10	1500
vp.p50	2500
vp.p90	4230
acoustic_impedance_MRayl.min	2.364
acoustic_impedance_MRayl.max	10.81
acoustic_impedance_MRayl.mean	6.145
vs.min	0
vs.max	2802
vs.mean	1163
vs.std	803.3
vs.p10	0
vs.p50	1058
vs.p90	2414
vp_vs_ratio.mean	2.489
vp_vs_ratio.std	0.774
vp_vs_ratio.p10	1.731
vp_vs_ratio.p90	3.726
poisson_ratio.mean	0.3653
poisson_ratio.p10	0.2494
poisson_ratio.p90	0.4612

Generated Figures

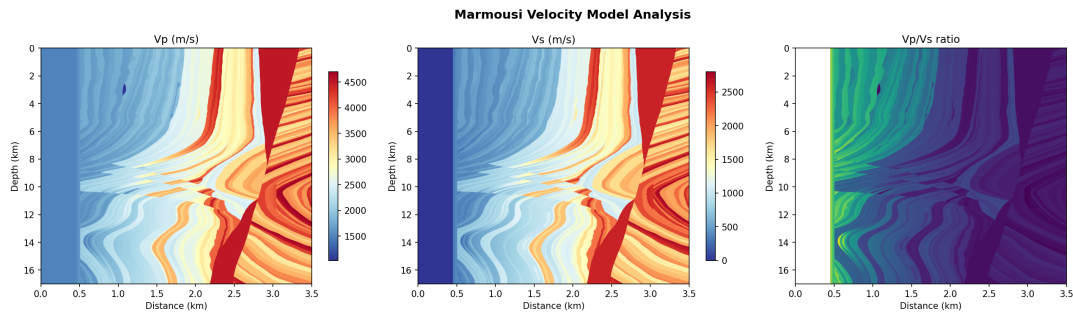


Figure 1: Velocity Model 20260501 115601

Geophysical Analysis & Interpretation

Technical Summary: Analysis of Marmousi Synthetic Dataset

This report summarizes the automated petrophysical characterization of the Marmousi synthetic velocity model, focusing on parameters relevant for porosity prediction. The analysis provides key statistical insights into the model's acoustic and elastic properties, offering a foundation for further geological and reservoir interpretation.

1. Velocity Statistics and Petrophysical Context

The compressional wave velocity (V_p) within the Marmousi model exhibits a substantial range, spanning from values indicative of unconsolidated near-surface sediments to those characteristic of more competent, consolidated lithologies or deeper formations. The broad distribution, with a significant standard deviation, underscores the inherent geological heterogeneity typical of thrust-and-fold belts. Similarly, shear wave velocity (V_s) also presents a wide distribution, reflecting variations in rock stiffness and saturation. A critical observation is the presence of V_s values at or near zero. Such extremely low V_s values commonly signify zones where shear waves are highly attenuated or do not propagate, often associated with highly unconsolidated, fluid-filled sediments (e.g., water-saturated sands or muds) or potentially gas-bearing zones. These low V_s regions are paramount for porosity prediction, as they frequently correlate with high pore fluid content and lower rock frame stiffness. The substantial standard deviations for both V_p and V_s corroborate the complex geological nature of the Marmousi model, implying a mixture of lithologies, compaction states, and potentially varying fluid types throughout the dataset.

2. V_p/V_s Ratio and Poisson's Ratio Implications

The derived V_p/V_s ratio and Poisson's ratio statistics provide crucial insights into the model's lithology and fluid content. The average V_p/V_s ratio, along with the average Poisson's ratio, falls within a range typically associated with shaly lithologies or water-saturated sediments. The upper percentiles for both ratios are notably high, strongly suggesting the presence of significant volumes of unconsolidated, highly porous, and fluid-saturated formations, consistent with the low V_s values observed. Conversely, the lower percentiles indicate regions of lower V_p/V_s and Poisson's ratio, potentially representing

more consolidated sandstones, tighter carbonates, or even hydrocarbon-bearing zones, where the rock frame is stiffer. The exceptionally wide range in these elastic properties implies a highly variable geological environment with diverse pore fluid characteristics and rock mechanical properties. These elastic attributes are directly sensitive to fluid substitution effects and lithological variations, making them fundamental for distinguishing between different reservoir types and fluid phases.

3. Forward Modelling Outcome and Synthetic Data Quality

The provided analysis focuses on characterizing the inherent properties of the pre-existing Marmousi synthetic velocity model. Therefore, this pipeline does not present an outcome of *new* forward modeling. Instead, the generated velocity model figure serves as a visualization of the input model, which is widely recognized in the geophysical community for its detailed and realistic representation of complex subsurface structures and strong lateral heterogeneity. The quality of this synthetic dataset is generally considered high, providing a robust platform for testing and validating geophysical workflows, particularly those aimed at challenging structural settings and petrophysical inversions. The internal consistency of the derived parameters (acoustic impedance, elastic ratios) within the analysis pipeline itself indicates a sound characterization of the input model.

4. FWI Convergence and Velocity Error Metrics

The current analytical output does not include metrics pertaining to Full Waveform Inversion (FWI) convergence or velocity model error. This suggests that the pipeline's primary function, in this instance, is the petrophysical characterization of a given velocity model, rather than the assessment of a velocity model building process. For subsequent phases of exploration, if the velocity model were derived through FWI, these metrics would be essential for evaluating the accuracy and reliability of the velocity field.

5. Geophysical Conclusions

The detailed petrophysical characterization of the Marmousi synthetic dataset reveals a geologically complex environment with significant variations in lithology, compaction, and pore fluid content. The prevalence of high V_p/V_s and Poisson's ratio values, coupled with extremely low shear wave velocities in certain regions, strongly indicates the presence of highly porous, fluid-saturated zones critical for porosity prediction.

Actionable Geophysical Conclusions:

- **Investigate Low V_s Zones:** Detailed analysis of areas with extremely low or zero V_s values is crucial. These zones may represent highly porous, fluid-saturated (e.g., water or gas) units and warrant focused petrophysical investigation.
- **Calibrate Petrophysical Transforms:** Develop and calibrate robust petrophysical transforms for porosity prediction that account for the wide range of V_p/V_s and Poisson's ratio observed, particularly addressing the non-linearities associated with high fluid saturation and unconsolidated sediments.
- **Multi-Attribute Analysis:** Leverage the derived acoustic impedance, V_p/V_s ratio, and Poisson's ratio in a multi-attribute analysis framework to enhance the discrimination of lithology and fluid types, leading to more accurate porosity estimations.

- **Sensitivity Analysis:** Perform a sensitivity analysis to understand how variations in V_p , V_s , and density influence porosity predictions, especially in regions exhibiting extreme elastic properties.
- **Workflow Validation:** The Marmousi dataset's inherent complexity makes it an ideal benchmark for validating advanced porosity prediction workflows before application to real-world seismic data from similar geological settings.

Document Information

Generated automatically by the **Seismic Agent Multi-Agent System** on May 01, 2026. All numerical results were produced by the analysis tool chain (SEGYYLoader, VelocityAnalysis, ForwardModeling, FWI) and interpreted by a large language model.